

Singular value decomposition (SVD) is the generalization of the [Eigen Decomposition] to non-square matrices. Formally, for any matrix $A \in \mathbb{R}^{m \times n}$, the SVD is given by

$$A = U\Sigma V^T$$

where $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ are orthogonal matrices, and $\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r) \in \mathbb{R}^{m \times n}$, where $r = \min(m, n)$ and $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r \geq 0$ are the singular values of A . The columns of U are the left singular vectors, and the columns of V are the right singular vectors.

0.0.1 See also

0.0.2 References